

Time: 50min

Note: Attempt all questions.

MM: 15

- Assume that we want to design a neural network with two layers by composing two linear layers. That is, the output of the first layer becomes the input of the second layer. Why would such a naive composition not work? (2)
- Explain the following code:

```
df = pd.read_csv('mnist.csv') // Assume 70000 examples
Y = df.loc[:, '0']
X = df.drop(['0'], axis=1)
print(X.shape)
```

 (1)
- Explain the following code:

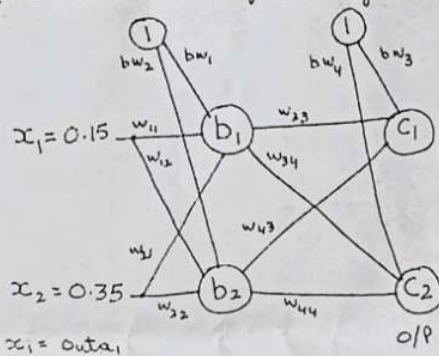
```
df = pd.DataFrame({
    'City': ['SFO', 'LA', 'SD'],
    'Team Name': ['Red', 'Blue', 'White'],
    'Wins': [10, 8, 5]
})
df.iloc[:, -1]
df.iloc[:, :-1]
```

 (1)
- What will be the output?

```
X_train = 0.5 * random_seed.randn(10, 2)
X_train = np.r_[X_train + 3, X_train]
X_train = pd.DataFrame(X_train, columns=["x", "y"])
```

 (1)
- What is an autoencoder? Give the equation along with dimensions. How does it help in unsupervised learning? (2)
- How do you obtain b_0 and b_1 for a given dataset for $y = b_0 + b_1 * x$? Explain clearly. (2)
- Solve the question given below: (6)

Ques: Consider the following neural network



$x_1 = \text{out}_{a1}$
 $x_2 = \text{out}_{a2}$

$w_{11} = 0.1$

$w_{12} = 0.2$

$w_{21} = 0.12$

$w_{22} = 0.17$

$bw_1 = 0.8$

$bw_2 = 0.25$

$w_{33} = 0.05$

$w_{34} = 0.40$

$w_{43} = 0.33$

$w_{44} = 0.07$

$bw_3 = 0.15$

$bw_4 = 0.70$

o/p Layer

Activation Functions

$$z = f(x) = \frac{1}{1 + e^{-x}}$$

For o/p Layer

$$SE = \frac{1}{2} \sum (t_i - z_i)^2$$

Desired

Actual Obtained

$$\therefore \delta_z = (z - t) z (1 - z)$$

Note: Output of $b_1 = \text{out}_{b1}$

Also, $\frac{\partial E}{\partial w}$ for node c_1, c_2 (Negatives are cancelled)

$$= (z - t) z (1 - z) \text{out}_{b1}$$

Ques

- Obtain values for $\frac{\partial E}{\partial w_{33}}, \frac{\partial E}{\partial w_{34}}, \frac{\partial E}{\partial w_{43}}, \frac{\partial E}{\partial w_{44}}$
- Find total error, SE

$$\frac{\partial E}{\partial w_{11}} = \left(\sum_c \delta_z w_{bc} \right) \text{out}_{b1} (1 - \text{out}_{b1}) \text{out}_{a1}$$